Josue Torres-Fonseca

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EDUCATION

Doctor of Philosophy in Computer Science and Engineering **Expected May 2028**
University of Michigan, Ann Arbor, MI Advisor: Dr. Joyce Chai GPA: 4.0

Master of Science in Computer Science and Engineering **Graduated May 2025**
University of Michigan, Ann Arbor, MI Advisor: Dr. Joyce Chai GPA: 4.0

Bachelor of Science in Computer Science, *Minor: Applied Mathematics* **Graduated May 2023**
Boise State University, Boise, ID Advisor: Dr. Casey Kennington GPA: 3.889, *Magna Cum Laude*

RESEARCH EXPERIENCE

PhD Candidate, *Situated Language and Embodied Dialogue Group* **Aug 2023 – Present**
University of Michigan, supervised by Dr. Joyce Chai Ann Arbor, MI

- **Tracing the Safety Signal in VLM-Planner → VLA-Controller Pipelines** *Ongoing*
  - Evaluating safety in hierarchical robot pipelines where a high-level VLM planner issues safety-aware instructions to a low-level VLA controller ($\pi_{0.5}$) on long-horizon BEHAVIOR-1K household tasks, with hazards drawn from real injury data (NEISS)
  - Diagnosing whether unsafe outcomes originate in the planner or the controller
  - Using mechanistic interpretability (activation patching, SAEs, linear probing) to locate where the safety signal lives inside the VLM → pipeline and how it travels from VLM planner to VLA action expert
  - Developing methods that amplify safety features identified with sparse autoencoders (SAEs), preserving the safety signal as it passes from VLM planner to VLA controller during action prediction while maintaining task success; evaluated on held-out hazard categories
- **ReMI Longitudinal: Building Resilience through Multi-Week Robot Interaction** *Ongoing*
  - Extending ReMI into a 2-week in-home deployment that aims to strengthen resilience in individuals living with cognitive decline through resilience education, guided discussion, and mindfulness practice
- **ReMI: A Conversational Robot to Elicit Resilience Reflection** *HRI 2027, under review*
  - Designed and built an embodied conversational robot that encourages individuals living with cognitive decline to reflect on their resilience, co-designed with 10 dementia experts and 14 people with cognitive impairment
  - Developed a structured dialogue manager that steps through rapport-building, challenges, and resilience reflection phases; blinded raters scored its pacing above end-to-end LLM prompting in 35 of 40 simulated conversations
  - Fine-tuned Qwen3-VL-32B on 15K dialogue turns distilled from Gemini 2.5 Pro, plus a resilience topic classifier reaching 84% accuracy; served both on-premise with vLLM to protect participant privacy
  - Deployed in 7 lab and 3 in-home studies with individuals living with cognitive decline; co-first author
- **SafetyALFRED: Evaluating Safety-Conscious Planning of Multimodal LLMs** *Findings of ACL 2026*
  - Created a safety-centric dataset using PDDL to generate trajectories that complete household tasks while mitigating six categories of kitchen hazards across 30 AI2-THOR scenes
  - Evaluated 11 state-of-the-art VLMs on hazard recognition and safe embodied planning, finding models correctly recognized hazards up to 92% of the time but fixed them less than 60% of the time
  - Developed a decoupled multi-agent framework separating hazard recognition from task execution, raising the rate at which Qwen3-VL-32B fixed appliance-misuse hazards from 0.7% to 71.1%
  - Open-sourced code and dataset; presented a poster at ACL 2026

Undergraduate Researcher, *Speech Language and Interactive Machines Group* **Aug 2021 – Aug 2023**
Boise State University, supervised by Dr. Casey Kennington Boise, ID

- **Symbol and Communicative Grounding through Object Permanence on a Mobile Robot** *SIGdial 2022*
  - Designed and built an incremental spoken dialogue system capable of transcribing and understanding speech, detecting and gaining a concrete understanding of objects, and mapping objects using the Anki Cozmo robot
  - Approximated object permanence by extending Cozmo's SLAM map to remember each object it has seen and its location; in an A/B study with 24 participants, this raised the robot's accuracy answering questions about learned objects from 70.0% to 82.7% and nearly doubled the questions participants asked
- **HADREB: A Dataset of Human Appraisals of Robot Emotional Behaviors** *LREC 2022*

- Collected and analyzed a dataset of over 1,000 robot emotional behaviors described and rated by 32 participants on the Misty II and Cozmo robots
- Showed a zero-shot language model predicts perceived emotion from the descriptions with up to 79.3% accuracy

Undergraduate Researcher, *Language & Intelligence Group* **May 2022 – Nov 2022**
Massachusetts Institute of Technology, supervised by Dr. Jacob Andreas Cambridge, MA
MIT Summer Research Program (MSRP)

- Designed and built an interpretable ML model providing neuron annotations using no data
- Presented a poster and delivered a lightning talk at the end of the MIT Summer Research Program

Undergraduate Researcher, *Intelligent Security Group* **Jun 2020 – Aug 2020**
Texas State University, supervised by Dr. Mina Guirguis San Marcos, TX
NSF Research Experiences for Undergraduates (REU), Smart & Connected Communities

- Designed and built a game and reward system to secure a simulated water tank using game theory
- Implemented check blocks acting as both an intrusion detection and prevention system for the simulated tank

INDUSTRY EXPERIENCE

Software Engineer Intern **May 2021 – Aug 2021**
Paylocity Meridian, ID

- Designed, developed, tested, and deployed new features in project releases as part of a 13-member team, working with product owners and leads to translate use cases into features
- Applied Agile and Scrum; gained full-stack experience with JavaScript, React/Redux, C#, .NET, SQL, and Git

PUBLICATIONS

Robot to Elicit Resilience Reflection for Individuals Living with Cognitive Decline **Under Review**
Long-Jing Hsu, Josue Torres-Fonseca*, Janice Bays, Joyce Chai, and Patricia Alves-Oliveira. 2027. Robot to Elicit Resilience Reflection for Individuals Living with Cognitive Decline. Under review at the ACM/IEEE International Conference on Human-Robot Interaction (HRI 2027).* **(*Equal contribution, co-first authors)**

SafetyALFRED: Evaluating Safety-Conscious Planning of Multimodal Large Language Models **Jul 2026**
Josue Torres-Fonseca, Naihao Deng, Yinpei Dai, Shane Storks, Yichi Zhang, Rada Mihalcea, Casey Kennington, Joyce Chai. 2026. SafetyALFRED: Evaluating Safety-Conscious Planning of Multimodal Large Language Models. In Findings of the Association for Computational Linguistics: ACL 2026, pages 37190–37211, San Diego, CA, USA. Association for Computational Linguistics. arXiv:2604.19638

The Mechanistic Emergence of Symbol Grounding in Language Models **Jul 2026**
Shuyu Wu, Ziqiao Ma, Xiaoxi Luo, Yidong Huang, Josue Torres-Fonseca, Freda Shi, Joyce Chai. 2026. The Mechanistic Emergence of Symbol Grounding in Language Models. In Proceedings of the International Conference on Machine Learning (ICML) 2026, Seoul, South Korea. arXiv:2510.13796

rrSDS 2.0: Incremental, Modular, Distributed, Multimodal Spoken Dialogue with Robotic Platforms **Aug 2025**
Anna Manaseryan, Porter Rigby, Brooke Matthews, Catherine Henry, Josue Torres-Fonseca, Ryan Whetten, Enoch Levandovsky, and Casey Kennington. 2025. rrSDS 2.0: Incremental, Modular, Distributed, Multimodal Spoken Dialogue with Robotic Platforms. In Proceedings of the 26th Annual Meeting of the Special Interest Group on Discourse and Dialogue, pages 637–640, Avignon, France. Association for Computational Linguistics.

Symbol and Communicative Grounding through Object Permanence with a Mobile Robot **Sep 2022**
Josue Torres-Fonseca, Catherine Henry, and Casey Kennington. 2022. Symbol and Communicative Grounding through Object Permanence with a Mobile Robot. In Proceedings of the 23rd Annual Meeting of the Special Interest Group on Discourse and Dialogue, pages 124–134, Edinburgh, UK. Association for Computational Linguistics.

HADREB: Human Appraisals and (English) Descriptions of Robot Emotional Behaviors **Jun 2022**
Josue Torres-Fonseca and Casey Kennington. 2022. HADREB: Human Appraisals and (English) Descriptions of Robot Emotional Behaviors. In Proceedings of the Thirteenth Language Resources and Evaluation Conference, pages 5739–5748, Marseille, France. European Language Resources Association.

TALKS & PRESENTATIONS

SafetyALFRED, poster, *ACL 2026*, San Diego, CA Jul 2026
Grounding through Object Permanence, talk and poster, *SIGdial 2022*, Edinburgh, UK Sep 2022

Data-Free Neuron Annotation, poster and lightning talk, *MIT Summer Research Program*, Cambridge, MA Aug 2022
HADREB, video presentation, *LREC 2022*, Marseille, France Jun 2022

HONORS & AWARDS

Gemini Academic Program Award	**($10,000)**	**Spring 2026 – Spring 2027**
NSF Graduate Research Fellowship	**($159,000)**	**Fall 2023 – Spring 2027**
Boise State University Top Ten Scholar		**Spring 2023**
Barry M. Goldwater Scholarship	**($7,500)**	**Fall 2022 – Spring 2023**
Dean's List, Boise State University		**Fall 2018 – Spring 2023**
Federal Employee Education & Assistance Fund Scholarship	**($1,000)**	**Fall 2018**

TEACHING EXPERIENCE

Tutor, *Operating Systems*, Boise State University Jan 2023 – May 2023
Tutor, *Introduction to Theory of Computation*, Boise State University Jan 2021 – May 2021
Tutor, *Data Structures*, Boise State University Aug 2020 – Dec 2020
Lab Assistant, *Microprocessors*, Boise State University Aug 2020 – Dec 2020

ORGANIZATIONS

Treasurer, *Computer Science and Engineering Graduate Students*, University of Michigan Fall 2024 – Spring 2025

- Managed organization funds and helped organize events

President, *Artificial Intelligence Club* Fall 2024 – Spring 2025

- Ran professional and technical development workshops on Python, machine learning, and data science

President, *Competitive Cybersecurity and Cyber-Physical Club*, Boise State University Fall 2019 – Spring 2022

- Competed in the CyberForce Competition at Idaho National Laboratory (46th of 103 teams, Fall 2019)
- Presented on data scraping and reverse engineering with Ghidra; led Capture the Flag activities for members